

Day 3 - Selenium Locators and Web Interaction

Methods in Java

A method is a reusable block of code used to perform a specific task.

Advantages:

- Reusability
- Readability
- Reduced duplication
- Easier maintenance

Methods are heavily used in Selenium frameworks for actions like:

- Login
- Logout
- Search
- Browser Launch

Methods Example

```
class Demo {  
    static void greet() {  
        System.out.println("Welcome");  
    }  
    public static void main(String[] args) {  
        greet();  
    }  
}
```

Constructors

A constructor is a special method automatically called when an object is created.

Main Uses:

- Object initialization
- Setup operations
- Driver initialization in Selenium
- Page initialization

Constructor Example

```
class Student {  
    Student() {  
        System.out.println("Constructor Called");  
    }  
}
```

```
        public static void main(String[] args) {  
            Student s = new Student();  
        }  
    }
```

this Keyword

The this keyword refers to the current object of the class.

Used to differentiate:

- Instance variables
- Local variables

Widely used in Selenium frameworks and Page Object Model.

this Keyword Example

```
class Employee {  
    int id;  
    Employee(int id) {  
        this.id = id;  
    }  
    void display() {  
        System.out.println(id);  
    }  
    public static void main(String[] args) {  
        Employee e = new Employee(101);  
        e.display();  
    }  
}
```

Selenium Locators

Locators are used to identify web elements on a webpage.

Examples:

- Search box
- Login button
- Password field
- Checkbox
- Dropdown

Types of Locators:

- id
- name
- className
- xpath
- cssSelector

findElement() Method

The `findElement()` method is used to locate web elements.

Syntax:
`driver.findElement()`

Structure:
`driver.findElement(By.id("value"));`

ID Locator

The id locator is the fastest and most preferred locator.

Example HTML:
`<input id="username">`

Selenium Code:
`driver.findElement(By.id("username"));`

Name Locator

The name locator identifies elements using the name attribute.

Example HTML:
`<input name="email">`

Selenium Code:
`driver.findElement(By.name("email"));`

className Locator

The className locator identifies elements using CSS class names.

Example HTML:
`<input class="search-box">`

Selenium Code:
`driver.findElement(By.className("search-box"));`

sendKeys()

The `sendKeys()` method is used to type text into input fields.

Example:
`driver.findElement(By.name("q"))
 .sendKeys("Selenium");`

click()

The `click()` method is used to click buttons, links, checkboxes, and other clickable elements.

Example:
`driver.findElement(By.id("login"))
 .click();`

Google Search Automation

```

import org.openqa.selenium.By;
import org.openqa.selenium.Keys;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class GoogleSearch {

    public static void main(String[] args) throws InterruptedException {

        WebDriver driver = new ChromeDriver();

        driver.manage().window().maximize();

        driver.get("https://www.google.com");

        Thread.sleep(2000);

        driver.findElement(By.name("q"))
            .sendKeys("Selenium WebDriver");

        Thread.sleep(2000);

        driver.findElement(By.name("q"))
            .sendKeys(Keys.ENTER);

        Thread.sleep(3000);

        driver.quit();

    }
}

```

Program Flow

```

Open Chrome Browser
↓
Open Google
↓
Find Search Box
↓
Type Selenium WebDriver
↓
Press ENTER
↓
Display Search Results
↓
Close Browser

```

Keys.ENTER

Keys.ENTER simulates pressing the ENTER key from the keyboard.

Examples of keyboard actions:

- ENTER
- TAB
- ESCAPE
- SPACE

Inspecting Elements

Steps:

1. Open website
2. Right click on element
3. Click Inspect
4. HTML source opens

Example:

```
<input name="q">
```